

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Eight Semester of B. Tech (EE) Examination****May 2018****EE415 Advances in Power System****Date: 09.05.2018, Wednesday****Time: 01.30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1. Draw and explain different types of high voltage direct current (HVDC) links. Why bipolar links are more preferable? **[04]**

Q – 2. Answer the following questions (Any four) **[16]**

- (1) Explain the major components of a high voltage direct current (HVDC) transmission in converter station unit with necessary diagram.
- (2) Explain characteristics or features of Smart grid.
- (3) Draw and explain the actual control characteristics of high voltage direct current (HVDC) system.
- (4) State and justify the advantages and disadvantages of high voltage direct current (HVDC) transmission system.
- (5) Compare current source converter (CSC) and voltage source converter (VSC).

Q - 3 Solve the following examples (Any three) **[15]**

- (1) A back to back high voltage direct current (HVDC) link with one bridge at each end is transmitting 100 MW with $V_d=100$ kV. If $\alpha=15^\circ$, $\gamma=18^\circ$. Find V_{dor} , V_{doi} , Q_r and Q_i . Assume $R_{cr}=R_{ci}=12\Omega$. If the DC link is controlled such that Q_i is kept at value calculated earlier, find V_d , I_d , Q_r , α and γ for $P_d= 50$ MW.
- (2) A high voltage direct current (HVDC) link delivers DC power at 500 kV at the inverter end with constant current controller at rectifier end set at 1000A. The equivalent resistances of the rectifier and inverter stations are 15Ω and 18Ω respectively. The DC resistance of the line is 20Ω . If the AC voltage at the rectifier is 400 kV. Find the percentage of tap changer required to maintain the current constant in the DC link without gate control. Also find the value of delay angle of the rectifier to maintain the above current in the absence of a tap changer.

- (3) A high voltage direct current (HVDC) link delivers DC power with AC line voltage to the rectifier being 230 kV and that at the inverter being 220 kV. The current order of the rectifier is 1000 A and that of the inverter is 900. Estimate the delay/ extinction angle of the rectifier/inverter when the DC voltage at the inverter end is 250 kV. Assume the DC resistance of the line as $20\ \Omega$.
- (4) A high voltage direct current (HVDC) link has the following parameters-AC line voltage at rectifier terminals 320 kV when delivering load of 500 MW at 335 KV. The inverter operator with $\gamma=21.5^\circ$, $X_{\text{rect}}= 58\ \Omega$, $X_{\text{inv}}=57\Omega$, $R_{\text{line}}=3\Omega$. Calculate (a) α , (b) AC line current and P.F. at the rectifier end, and (c) AC line current, voltage, P.F. at the inverter end.

SECTION – II

- Q - 4 Define the term Flexible Alternating Current Transmission System (FACTS). Also give the detail classification of different types of FACTS controllers. [05]
- Q – 5.a Explain the working principle of unified power flow controller (UPFC) with necessary diagram. [05]

OR

- Q – 5.a Derive the equation of active power flow of Quadrature Booster. [05]
- Q – 5.b Explain the working principle and V-I characteristics of Thyristor Controlled Series Compensator (TCSC). [05]
- Q – 5.c What are reasons for adopting distribution generation? [05]

OR

- Q – 5.c Write down the advantages and disadvantages of distributed generation. [05]
- Q – 6. Answer the following questions (Any three) [15]

- (1) Explain the working principle and V-I characteristics of Static VAR Compensator (SVC).
- (2) Draw single line diagram of Thyristor controlled phase angle regulator (TCPAR) and explain its operating principle and vector diagram.
- (3) Explain the working principle and V-I characteristics of Static Synchronous Compensator (STATCOM) with suitable vector diagram.
- (4) Explain voltage regulation and Line impedance compensation by unified power flow controller (UPFC).
